Developer technology trends

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IBM Data Analyst Capstone



OUTLINE



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EXECUTIVE SUMMARY



65,437 unique respondents remain after removing 20 duplicate rows.

JavaScript leads current language use at 57.3%. Python leads future interest at 38.3%.

PostgreSQL leads both database rankings.

These preferences describe a voluntary survey sample.

INTRODUCTION



Purpose

Compare current technology use with developers' stated future interests.

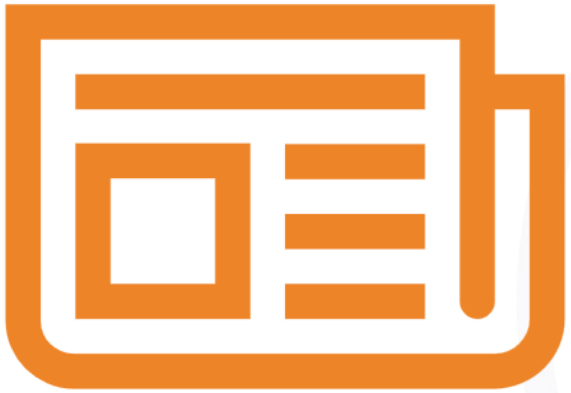
Audience

Data analysts and teams planning skills development.

Value

Identify common technologies and compare preferences with the separate job and salary exercises.

METHODOLOGY



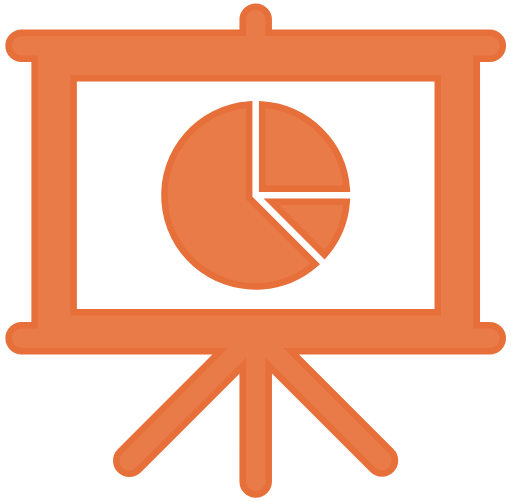
Collect the course jobs JSON through a local REST API and extract the salary HTML table.

Load 65,457 survey rows across 114 fields. Remove 20 exact duplicates.

Keep unanswered fields missing. Count each technology once per respondent.

Use positive compensation responses for the median. Apply an IQR filter only to its histogram.

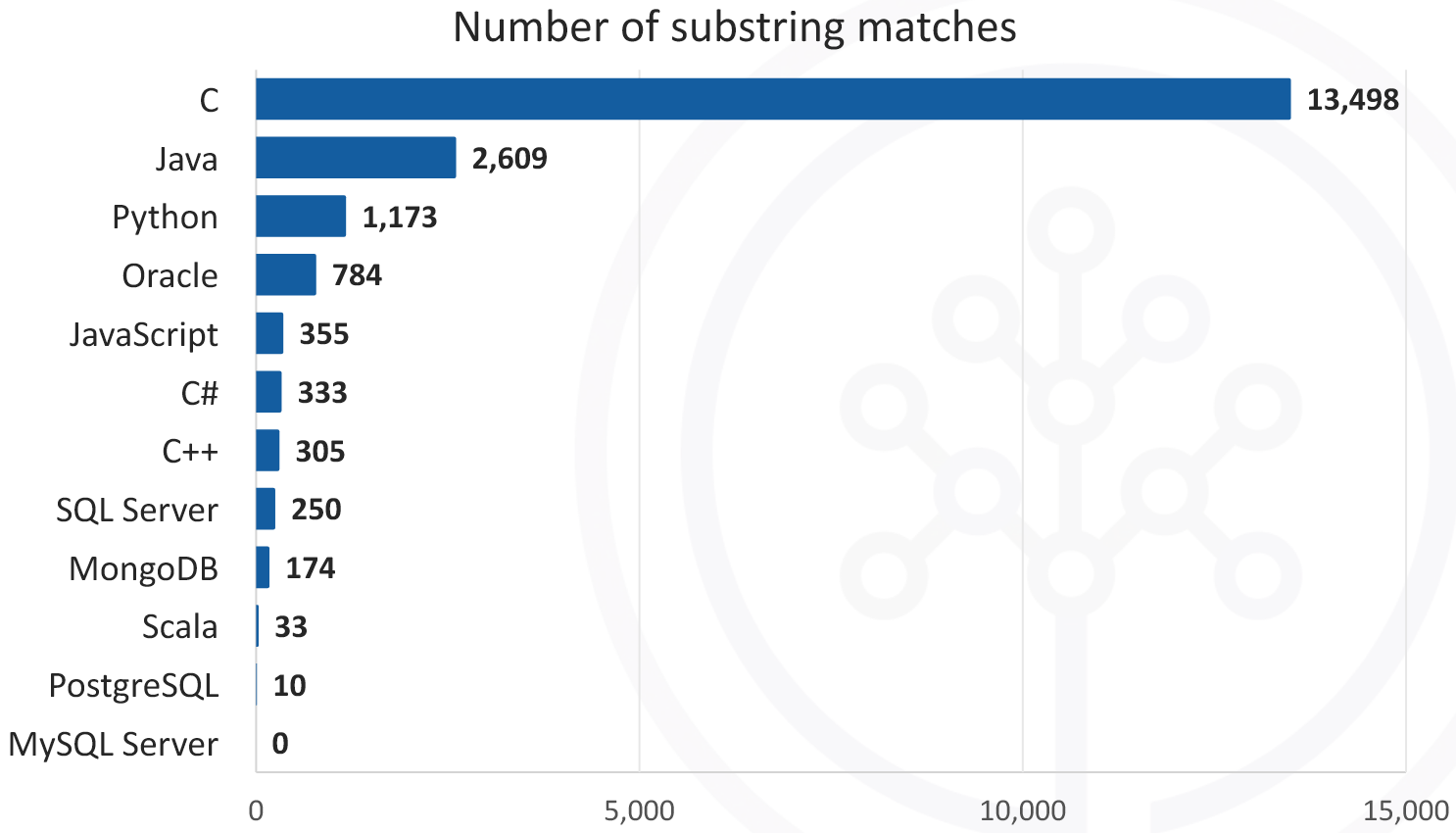
RESULTS



Collection exercises and technology preferences

27,005 modified job records
10 rows in the salary exercise
65,437 unique survey respondents

JOB POSTINGS



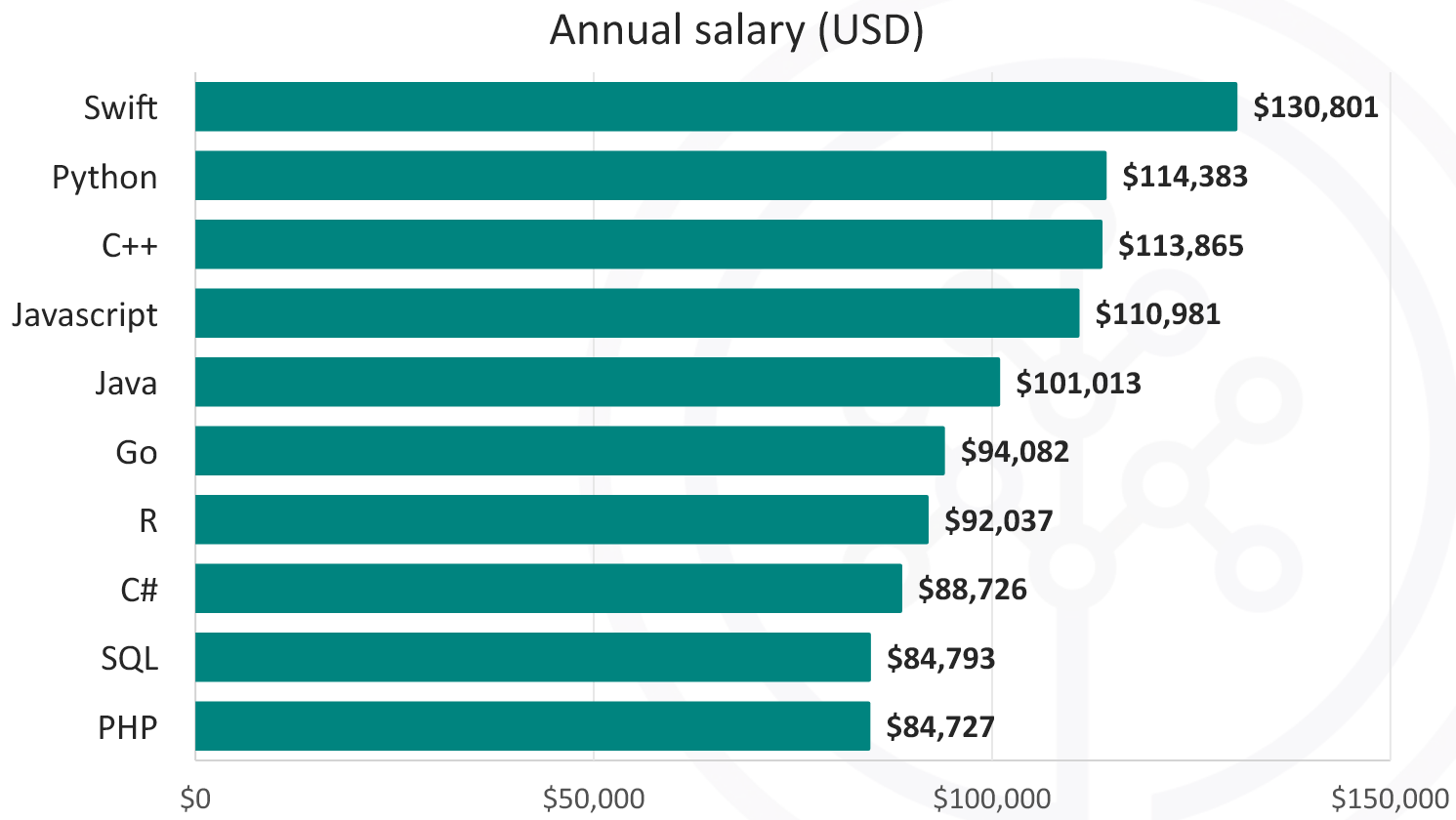
Course API matching

C: 13,498 matches
Java: 2,609
Python: 1,173

Substring matching can combine distinct technologies. The CSV also provides a bounded comparison.

Source: modified course jobs dataset, 27,005 records. Counts reflect course API substring matching.

POPULAR LANGUAGES



Salary HTML exercise

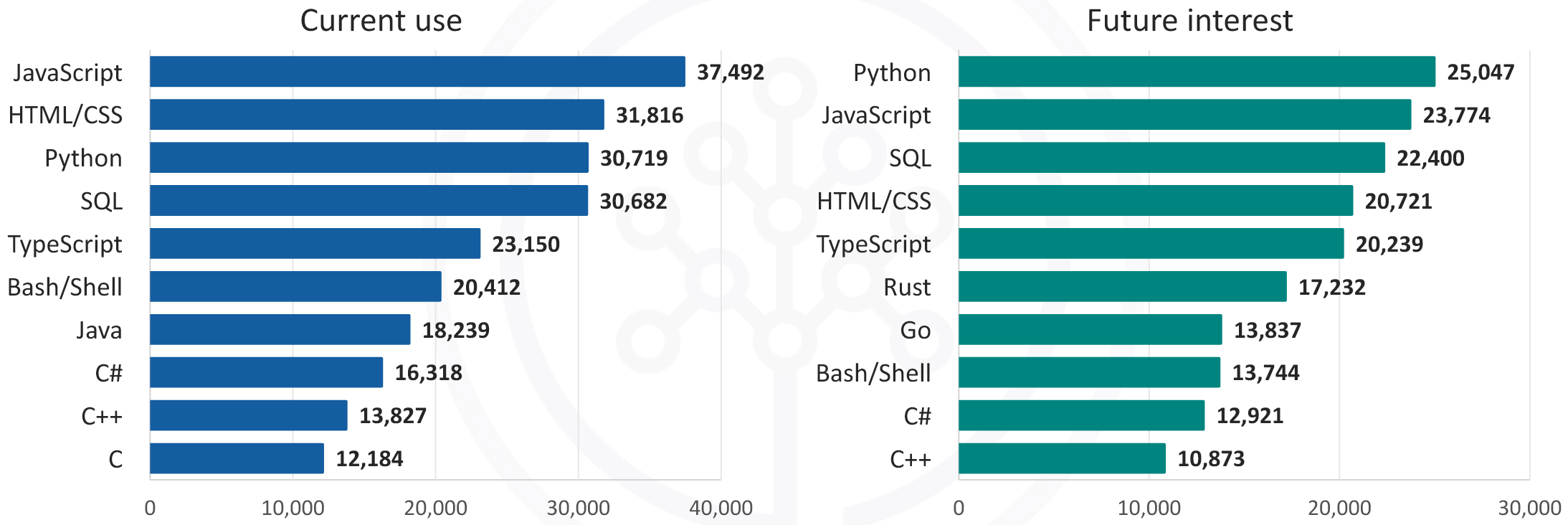
Swift: $130,801
Python: $114,383
C++: $113,865

The supplied table omits its sample size, survey date and methodology.

Source: supplied HTML salary table. Exercise values in USD, separate from survey compensation.

PROGRAMMING LANGUAGE TRENDS

JavaScript leads current use. Python leads stated future interest.



Respondent counts, top ten in each ranking. Multiple selections allowed. Unique respondents: 65,437.

LANGUAGE FINDINGS & IMPLICATIONS

Findings

JavaScript: 37,492 current users (57.3%).

Python: 25,047 future-interest mentions (38.3%).

SQL ranks among the four most common choices in both questions.

Implications

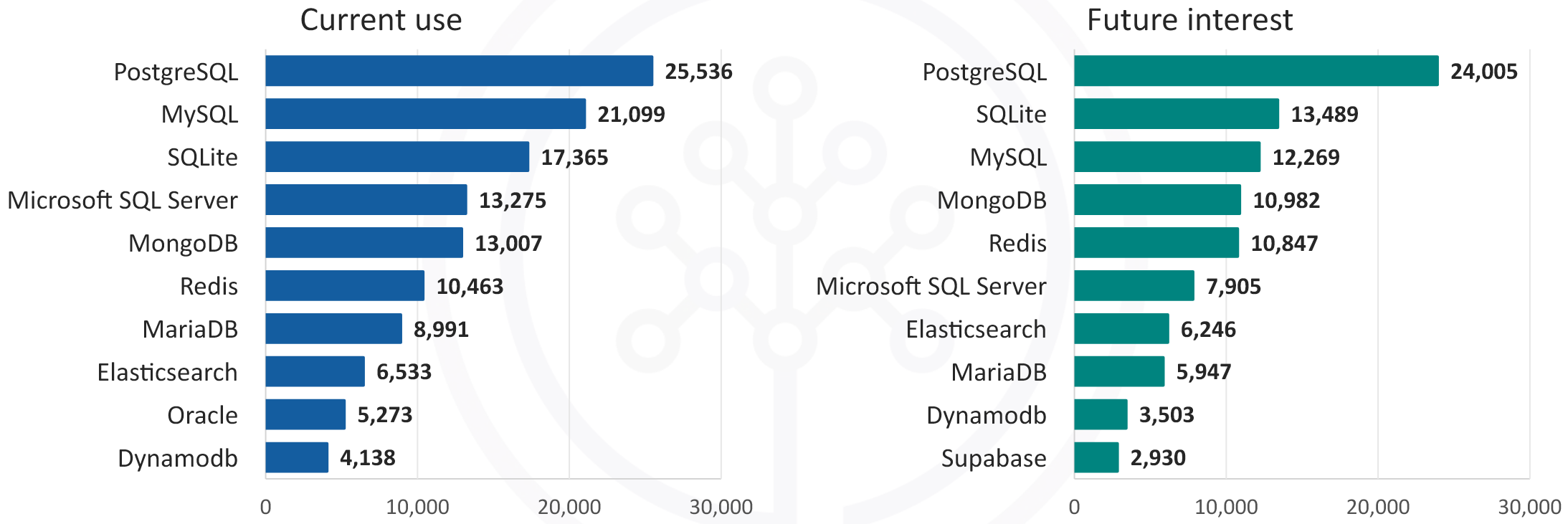
Python and SQL provide relevant foundations for data analysis.

JavaScript remains relevant to web work.

Compare preferences with actual team needs before choosing a learning plan.

DATABASE TRENDS

PostgreSQL leads current use and future interest.



Respondent counts, top ten in each ranking. Multiple selections allowed. Unique respondents: 65,437.

DATABASE FINDINGS & IMPLICATIONS

Findings

PostgreSQL: 25,536 current users and 24,005 future-interest mentions.

SQLite ranks second in future interest.

Redis interest exceeds current use by 384 mentions.

Implications

SQL and relational database skills fit the most common preferences.

PostgreSQL offers a relevant practice environment.

Selection also depends on workload and operating requirements.

DASHBOARD



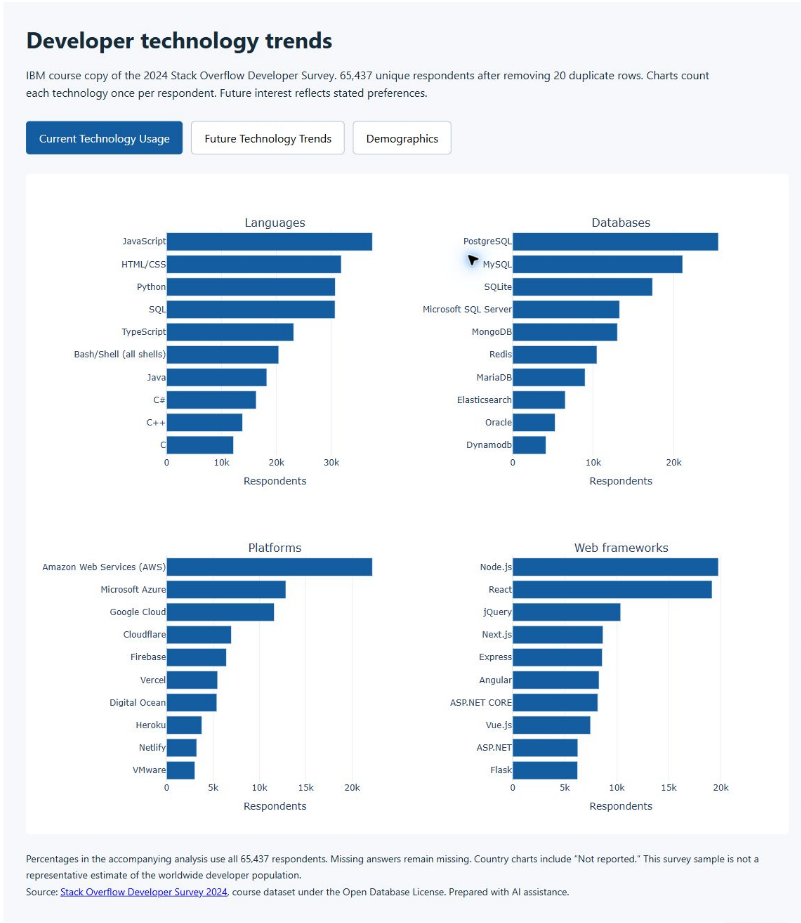
Three interactive portfolio views

Current Technology Usage
Future Technology Trends
Demographics

The accompanying HTML uses Plotly.

Portfolio dashboard. Course platform requirements and graded submission remain unverified.

CURRENT TECHNOLOGY USAGE



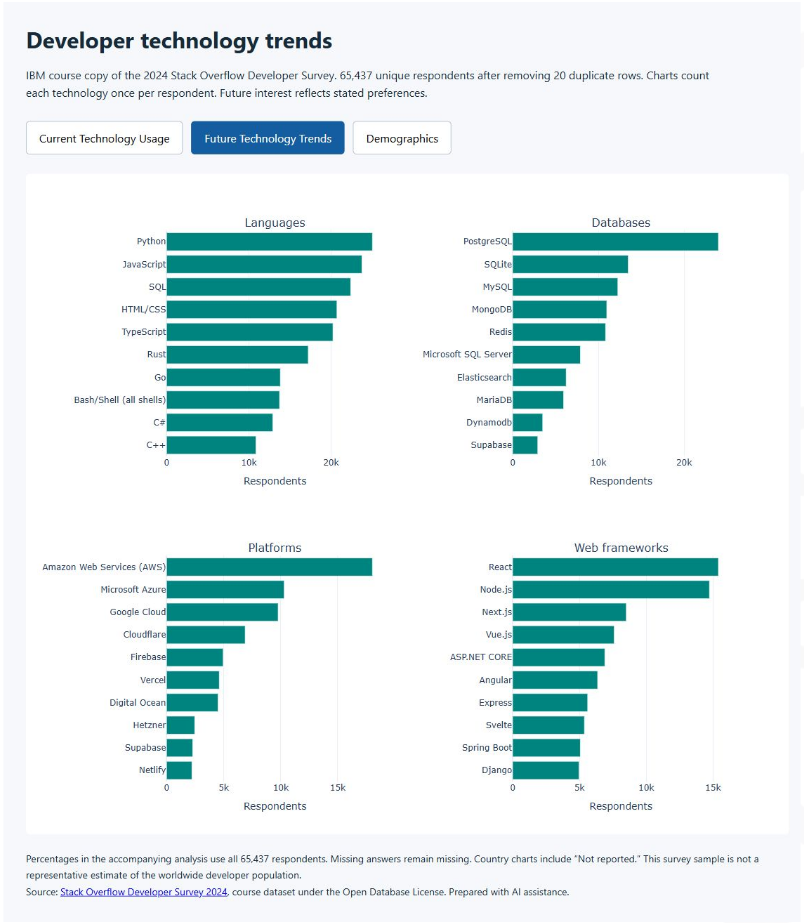
JavaScript leads languages with 37,492 mentions.

PostgreSQL leads databases with 25,536 mentions.

AWS leads platforms. Node.js leads current web frameworks.

Screenshot of the accompanying local Plotly dashboard. Interactive charts and full labels are in dashboard.html.

FUTURE TECHNOLOGY TRENDS



Python leads languages with 25,047 mentions.

PostgreSQL leads databases with 24,005 mentions.

React leads future interest among web frameworks.

Screenshot of the accompanying local Plotly dashboard. Interactive charts and full labels are in dashboard.html.

RESPONDENT DEMOGRAPHICS



The largest age band is 25–34 with 23,911 respondents (36.5%).

185 countries have reported responses. Another 6,507 respondents omit country.

Missing answers remain visible in the charts.

Screenshot of the accompanying local Plotly dashboard. Interactive charts and full labels are in dashboard.html.

DISCUSSION



Current use and future interest answer different questions in the same survey. Differences do not measure adoption over time.

Respondents select multiple technologies. Percentages therefore do not add to 100%.

Country and compensation coverage vary. Team planning needs context beyond these rankings.

OVERALL FINDINGS & IMPLICATIONS

Findings

Python, SQL and PostgreSQL appear prominently in developer preferences.

The 25–34 age band is the largest.

The modified jobs exercise depends strongly on its matching rule.

Implications

Prioritize relevant foundations, then adapt to the team's work.

Validate demand with a dated, representative jobs source.

Use consistent matching and disclose response coverage.

CONCLUSION



JavaScript leads current language use. Python leads future interest.

PostgreSQL leads both database rankings.

A reproducible pipeline links the notebook, aggregate tables and dashboard.

The results support descriptive comparisons within this survey sample.

APPENDIX: DATA QUALITY



Compensation coverage
23,435 positive annual values
Median: USD 65,000
42,002 missing values

The histogram excludes 978 positive IQR outliers only for display. The median includes every positive response.

Source links and calculation details appear in the accompanying notebook and README.

Analysis developed with AI assistance. Public source data and derived aggregates accompany the code.